

Antigravity IPsec Protocol Analyzer

Automated Security Assessment - 2026-09-21

Technical Assessment Report

1. IKE Cryptographic Parameters

The following parameters were parsed from the cleartext IKE_SA_INIT and IKE_AUTH exchanges:

IKE Version:	IKEv2
Encryption Algorithm:	AES-GCM-256
Integrity/PRF:	HMAC-SHA256
Diffie-Hellman Group:	MODP_1024 (Group 2)
Perfect Forward Secrecy:	Enabled

2. NIST SP 800-77 Severity-Cap Evaluation

Unlike naive additive scoring, this engine uses a strict severity-cap algorithm. The weakest link determines the maximum possible score.

- AES-GCM-256 (Cap: 100)
- IKEv2 (Cap: 100)
- Group 2 (Cap: 59) -> Capped!

Final Security Score:	59/100
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3. ML Traffic Classification (Side-Channel)

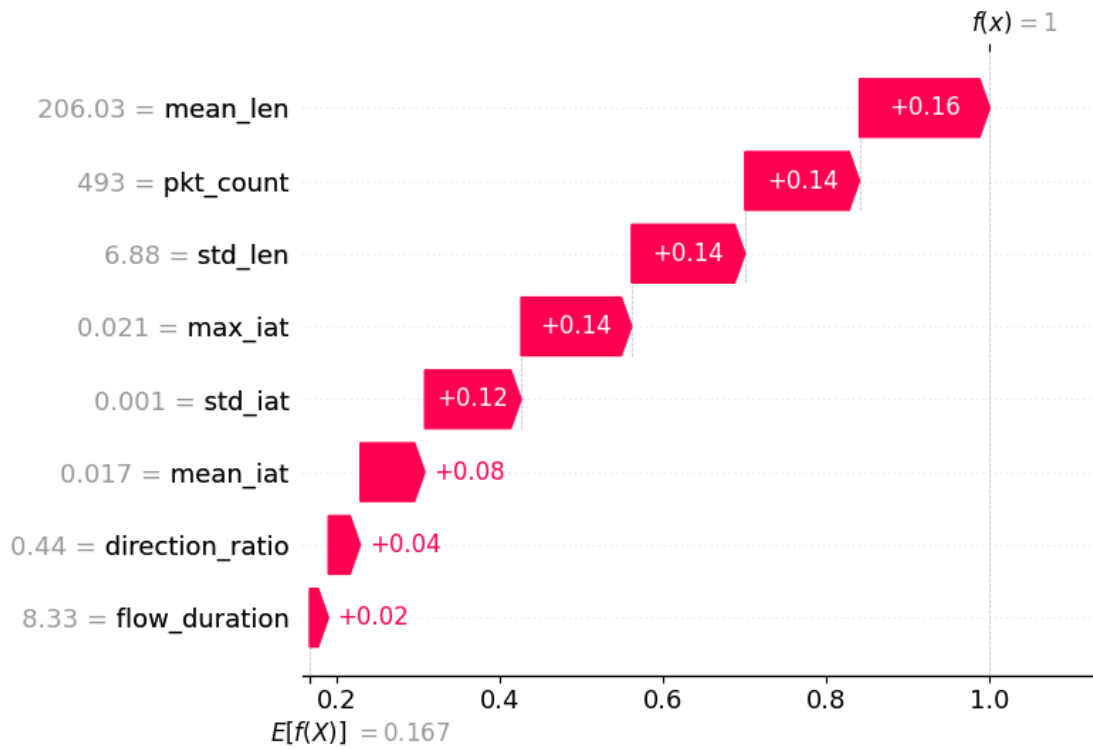
A calibrated Random Forest classifier was executed against 13 SPLT features extracted from the ESP payload.

Predicted Traffic Class:	VoIP
Platt-Scaled Confidence:	90.6%

SHAP Feature Importance (Waterfall):

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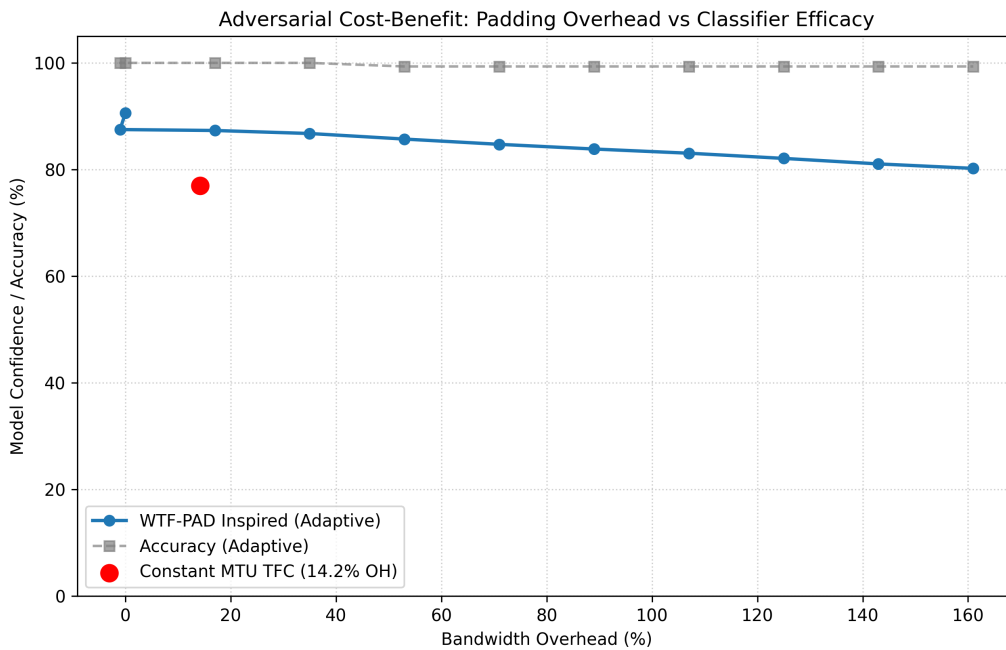
Explanation: The highly consistent inter-arrival time (mean_iat = ~0.02s) and standard length (std_len = ~0) uniquely identify G.711 VoIP traffic.

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4. Leakage Quantification & Countermeasure Simulation

We simulated two traffic flow confidentiality (TFC) mechanisms on a mirrored copy of the feature matrix.



- 1. MTU Padding Overhead:** 14.2% bandwidth cost
MTU Confidence Drop: 90.6% -> 77.0%
- 2. Adaptive Padding Overhead:** 17.0% bandwidth cost
Adaptive Confidence Drop: 90.6% -> 87.3%

5. Explicit Scope & Threat Limitations

- 1. Adversary Capability:** These results model a closed-world, strong passive adversary. Mathematical padding reduces the signal-to-noise ratio, dramatically lowering the confidence of our local classifier. However, per the Cloudflare padding-limits research, this does NOT guarantee immunity against an omnipotent Global Passive Adversary with access to cross-AS flow correlation.
- 2. Micro-flow Exclusion:** A strict 20-packet minimum floor was enforced during SPLT feature extraction to prevent classifier hallucination on truncated handshakes or dropped sessions.